



**Cambridge Instruments
& Engg. Co.**

EST. 1963 · HOWRAH, INDIA

EARTH TESTER · HAND-DRIVEN, METAL BODY

CIE/222M

Hand-driven analog earth tester — 3-terminal (earth resistance) and 4-terminal (earth resistance + soil resistivity). ±5% accuracy per IS: 9223/1979.

±5% ACCURACY

Analog Earth Tester — Metal Body

A hand-driven generator-type earth tester for direct measurement of earth resistance in power circuits, telecommunications, railway electrification, and industrial installations — available in 3-terminal (earth resistance) and 4-terminal (earth resistance plus soil resistivity) types, with single up to five-range configurations.

- **3-terminal or 4-terminal** — earth resistance, or + soil resistivity
- **±5% accuracy** — per IS: 9223/1979
- **Centrifugal governor** — constant voltage regardless of crank speed
- **Anti-magnetic aluminium body** — protected metallic flap over scale
- **Collapsible generator handle** — folds flush into casing
- **Single to five-range** — configurations available

Supplied as base instrument only — testing kit and carrying case available at extra cost.

Manufactured in India since 1963 — trusted by various State Electricity Boards, Indian Railways, cement and metal industries, and other sectors across India.



- V **2kV Dielectric Test** 1 min, circuit to containing case
- ± **±5% Accuracy** Per IS: 9223/1979
- B **Battery-Free** Hand-cranked generator

01 Full Specification Sheet

PARAMETER	SPECIFICATION
Body	Anti-magnetic aluminium
Generator Type	Hand-driven A.C. generator, centrifugal governor
Terminal Types	3-terminal (earth resistance) or 4-terminal (+ soil resistivity)
Scale Length	80 mm (approx.)
Accuracy	±5% (as per IS: 9223/1979)
Rotation Speed	160 r.p.m.
Single Range Options	0-1Ω to 0-5Ω · 0-10Ω to 0-100Ω · 0-200Ω to 0-10000Ω
Dual Range Options	0-1/10 · 0-2/10 · 0-3/30 · 0-10/100 Ω
Triple Range Options	0-1/10/100 · 0-2/10/100 · 0-2/20/200 · 0-3/30/300 · 0-10/100/1000 Ω
Four Range Options	0-1/10/100/1000 · 0-2/10/100/1000 · 0-10/100/1000/10000 Ω
Five Range Options	0-1/10/100/1000/10000 · 0-2/10/100/1000/10000 Ω
High Voltage Test	2 KV A.C. (r.m.s.), 1 min., circuit to containing case
Insulation Resistance Test	Better than 25 MΩ at 500V D.C.
Dimensions (L×W×H)	110 × 128 × 190 mm (approx.)
Weight	2.5 kg (approx.)
Warranty	12 months against manufacturing defects

Any configuration apart from the standard range & terminal type above can be built to order.

02 Typical Applications

- Earth resistance measurement for power circuits and lightning protection systems
- Telecommunications and railway electrification earthing checks
- Domestic and industrial electrical installation earthing
- 4-terminal type: soil resistivity surveys for civil engineering projects (tunnels, drainage, sewerage)

03 Optional Accessories

- + Testing kit — 3-terminal (spikes, wire, hammer, bag)
- + Testing kit — 4-terminal (spikes, wire, hammer, bag)
- + Carrying case
- + Any other range — on request

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Instruments for accuracy and
reliability since 1963.